Double-Lined from the Archive: Independent Dynamical Masses of M Dwarfs in Eclipsing Binaries from Public HARPS and ESPRESSO Spectra

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Abstract

An eclipsing binary of a solar-type star and an M dwarf gives model-independent masses for both stars once the M dwarf's own lines are measured, at optical flux ratios near 10^{-3}. This paper measures them from public ESO archive spectra with an analysis built independently of the published ones. The spectra are placed in the primary's frame at their measured velocities and divided by their series' median, cross-correlated with an empirical M1 V spectrum, cleaned of the star's and the sky's patterns and of five principal components, and summed along the companion's track over its semi-amplitude K_2 and its systemic offset. Each summit is set against 10^4 draws of two nulls, the orbit's phase moved and whole nights shuffled, with a Gumbel law for the tail, and the 35 binaries of the survey are accounted for together. TOI-1338/BEBOP-1 (260 ESPRESSO spectra), EBLM J0310−31 (114 HARPS spectra) and HD 4875 (35 HARPS spectra) give false-alarm probabilities of 10^{-11}, 8×10^{-12} and 1×10^{-8} and K_2 = $77.27 \pm 0.10 \pm 0.15$, $88.12 \pm 0.18 \pm 0.34$ and $75.29 \pm 0.30 \pm 0.87\,\mathrm{km\,s^{-1}}$ (statistical, then systematic from the series' parts by date and its seasons), against published values of 77.16±0.48, 88.27 ± 0.46 and $75.13 \pm 0.16\,\mathrm{km\,s^{-1}}$; HD 4875's companion had been measured only in the near-infrared. The companion's lines sit +1.05 to $+1.25\,\mathrm{km\,s^{-1}}$ from the primary's systemic velocity. Among the 35 southern EBLM binaries with archived HARPS series, one further companion stands with a false-alarm probability of 1.5×10^{-7} across the whole survey: EBLM J2101−45, whose 75 spectra give $K_2 = 65.41 \pm 0.32 \pm 0.32\,\mathrm{km\,s^{-1}}$ and minimum masses of 1.415 and $0.552\,\mathrm{M_\odot}$, stable under changes of template, band and principal components and recovered when a companion is injected into the same spectra. J0649−27, J0825−67 and J1141−37 reach 3.3 to 3.9σ alone but 2.1 to 3.0σ across the survey, and remain candidates.

1 Introduction

Masses and radii of M dwarfs set the properties inferred for the planets that orbit them, and the empirical relations behind those inferences rest on stars whose masses are measured without models. Detached double-lined eclipsing binaries provide such masses, but few are known below the fully convective boundary near $0.35\,\mathrm{M_\odot}$ [1, 2], since an M dwarf paired with a solar-type star contributes about a thousandth of the optical light and its lines are lost in the primary's. The EBLM project has characterised hundreds of these single-lined eclipsing binaries [3, 4, 5], measuring the primary's spectroscopic orbit and the eclipses and taking the primary's mass or radius from stellar models. The BEBOP survey for circumbinary planets has added years of precise velocities of the same stars with HARPS and SOPHIE [6, 7, 8].

The secondary's lines can nonetheless be recovered. High-resolution cross-correlation, developed to detect molecules in exoplanet atmospheres [9, 10], sums a weak spectrum along a known orbit after the stationary spectrum has been removed. Sebastian et al. applied it to the circumbinary planet host TOI-1338/BEBOP-1 [11, 12] and detected its M dwarf at 11σ in 103 ESPRESSO

spectra, measuring $K_2 = 77.16 \pm 0.48\,\mathrm{km\,s^{-1}}$ and masses of 1.098 ± 0.017 and $0.307 \pm 0.003\,\mathrm{M_\odot}$ [13], with the Saltire model of the cross-correlation map [14]. The method has since been applied to Kepler-16 with SOPHIE [15], to BEBOP-3 [16] and to five more EBLM binaries, among them EBLM J0310−31 [17]. In the near-infrared, where the M dwarf is relatively brighter, the secondary's semi-amplitude has been measured with SPIRou [18], CRIRES+ [19] and NIRPS [20, 21].

These measurements share a small number of analysis designs, and their accuracy has been tested by splitting each dataset in time and by simulation [13, 17]. An analysis built independently, with a different template, a different removal of the primary's spectrum and a different measure of significance, tests that accuracy from outside. The public archives also hold more spectra of these stars than the published analyses used: the ESO archive holds 114 HARPS spectra of EBLM J0310−31 above a signal-to-noise ratio of 10, against the 71 of ref. [17], and 264 ESPRESSO spectra of TOI-1338, against 103.

This paper describes such an analysis, the binaries approach of the seti-tools package, which shares its spectra machinery with the package's planet-frame and free-orbit companion searches and their orbitlight release [22], and applies it to public products of the ESO Science Archive alone. Section 2 selects the targets and spectra and Section 3 describes the method, its statistic and its calibration. Section 4 compares its results for three systems with published dynamical masses, which test its accuracy, its uncertainties and its false-alarm probabilities. Section 5 applies it to the southern EBLM binaries with HARPS series and separates the one new measurement from the candidates, and Section 6 discusses what the comparison shows.

2 Targets and spectra

The targets are the single-lined spectroscopic binaries with archived high-resolution spectra. The Gaia DR3 non-single-star catalogue [23] gives 203 121 single-lined orbits (solution types SB1, SB1C and AstroSpectroSB1) for stars brighter than $G = 13$, and the SB9 catalogue [24] 3 426 single-lined orbits. Each was matched within 10 arcsec to the ESO archive's series of 20 or more spectra of one target from HARPS, ESPRESSO, NIRPS, UVES or FEROS, giving 140 series of 126 stars. A series is searched when its orbit has a period below 100 d covered three or more times, the series holds 30 or more spectra, and the mass function allows a companion of $0.07\,\mathrm{M_\odot}$ or more around a primary of $1\,\mathrm{M_\odot}$. That leaves 68 series. Among them are the HARPS series of 37 EBLM binaries, named here by their EBLM designations, which this paper searches, and TOI-1338's ESPRESSO series; the others belong to O and B stars, interacting binaries with compact companions, and solar-type binaries observed mostly with UVES or FEROS.

The spectra are the archive's Phase 3 one-dimensional products, with barycentric wavelengths and the pipeline's flux and error. For each series, every product with a signal-to-noise ratio above 10 is taken, up to 300 spread evenly over the series' span.

3 Method

3.1 The primary's velocities and orbit

Each spectrum is placed on a grid uniform in velocity at $1\,\mathrm{km\,s^{-1}}$ from 3800 to 7880 Å, divided by its running median over 501 samples, with the telluric bands removed. Its velocity is the peak of its cross-correlation over $\pm 200\,\mathrm{km\,s^{-1}}$ with the line depths of a G0 V template, one minus the median of EBLM J0310−31's 114 HARPS spectra placed at their velocities on its Gaia orbit, refined by a parabola through the three highest samples. The primary's orbit is fitted to these velocities in two

steps. A grid over period (within 0.4 per cent of the catalogue's, in steps that move the last epoch by a fiftieth of an orbit), phase (in hundredths), eccentricity (to 0.6) and argument of periastron (in 20° steps) is searched with K_1 and the systemic velocity γ solved linearly at each point, and the best point starts a least-squares fit with a soft loss in period, time of periastron, $\sqrt{e}\cos\omega$, $\sqrt{e}\sin\omega$, K_1 and γ. The grid makes the fit independent of errors in the catalogue orbit, which for TOI-1338 put the period 0.021 d long and ω 58° away.

3.2 Residuals in the primary's frame

Each spectrum is then moved into the primary's frame at its own measured velocity, placed on the same grid, divided by its running median and by the series' median spectrum, and a cubic in wavelength is removed. A residual shift of a spectrum against the median of $0.5\,\mathrm{km\,s^{-1}}$ or more is taken out before the division. The telluric bands and the sodium sky lines at their telescope wavelengths are masked. Each wavelength's scatter across the series, floored at a tenth of its median, weights the cross-correlation.

3.3 Cross-correlation with an M-dwarf template

The template is an empirical M1 V spectrum: the line depths, one minus the median, of the archived HARPS spectra of GJ 367 [25] in its own frame, negative values set to zero. Each residual is cross-correlated with it by inverse variance over $\pm300\,\mathrm{km\,s^{-1}}$ at $1\,\mathrm{km\,s^{-1}}$, the result normalised so that a residual of unit noise gives unit scatter. Each cross-correlation row then loses three patterns that are not the companion's: the star's own residual pattern, the median row fitted to each row within $40\,\mathrm{km\,s^{-1}}$ of the star and subtracted; the sky's pattern, the rows' median in the telluric frame fitted back to each row; and five principal components of the rows, which carry the shapes the star's pattern takes from night to night. Each row is weighted by the inverse square of its robust scatter more than $50\,\mathrm{km\,s^{-1}}$ from the star.

3.4 The sum along the companion's track

The companion's barycentric velocity at time t is

$$v_2(t) = \gamma + \Delta v_{\mathrm{sys}} - K_2\left[\cos(\nu+\omega) + e\cos\omega\right], \tag{1}$$

with ν, ω and e from the primary's fitted orbit and Δv_{sys} the offset of the companion's lines from the primary's systemic velocity. With $C_i(v)$ the cleaned cross-correlation of spectrum i at velocity v in the frame in which it was placed, $v_{\star,i}$ the velocity at which it was placed and w_i its weight, the trail's value is

$$S(K_2, \Delta v_{\mathrm{sys}}) = \frac{\sum_i w_i\, m_i\, C_i(v_2(t_i) - v_{\star,i})}{\sum_i w_i\, m_i}, \tag{2}$$

read by linear interpolation, where m_i is one when the reading lies more than $30\,\mathrm{km\,s^{-1}}$ from the primary's lines and more than $10\,\mathrm{km\,s^{-1}}$ from the row's telluric velocity and zero otherwise. S is defined where the readings number at least a fifth of the rows and at least ten. The map of S over K_2 from K_1 to $160\,\mathrm{km\,s^{-1}}$ and over Δv_{sys} from -30 to $+30\,\mathrm{km\,s^{-1}}$, both at $0.5\,\mathrm{km\,s^{-1}}$, gives the peak within $5\,\mathrm{km\,s^{-1}}$ of zero offset, refined on grids of 0.1 and then $0.05\,\mathrm{km\,s^{-1}}$.

3.5 The statistic and its false-alarm probability

The search statistic is the summit $\hat{S} = \max_{K_2} S$ over a grid from K_1 to $250\,\mathrm{km\,s^{-1}}$ in steps of $0.25\,\mathrm{km\,s^{-1}}$, either at a fixed offset or, with the offset free, over the offsets within $5\,\mathrm{km\,s^{-1}}$ of zero

in steps of $0.5\,\mathrm{km\,s^{-1}}$ as well. A null draw computes the same summit with the companion's track replaced, by the orbit's phase moved by a uniform 0.1 to 0.9 of a period or by whole nights moved to other nights' dates, each night keeping its spectra's offsets from its start, so that what is tied to the orbital phase and what is tied to the hour or the night both stay in a null. Each binary is set against 10^4 draws of each null, $\hat{S}^{(j)}$, and three numbers describe its summit. The first is its height in units of the nulls' scatter,

$$z = \frac{\hat{S} - \langle \hat{S}^{(j)} \rangle}{\mathrm{sd}(\hat{S}^{(j)})}, \tag{3}$$

a standardised statistic rather than a probability. The second is the empirical false-alarm probability, $p_{\mathrm{emp}} = (1 + n_{\geq})/(10^4 + 1)$, with $n_{\geq}$ the draws at or above $\hat{S}$, which cannot fall below 10^{-4}. The third extends the nulls with a tail model. Each null summit is the largest of many correlated values of the trail, so the summits follow an extreme-value law, and the Gumbel law [27] with the nulls' mean and scatter gives

$$p_{\mathrm{G}} = 1 - \exp\left\{ -\exp\left[-\frac{\hat{S} - \mu}{\beta} \right] \right\}, \qquad \beta = \frac{\sqrt{6}}{\pi}\,\mathrm{sd}(\hat{S}^{(j)}), \quad \mu = \langle \hat{S}^{(j)} \rangle - 0.5772\,\beta. \tag{4}$$

The null of the two with the larger p_{G} is adopted, and p_{G} is quoted with the one-sided Gaussian significance of the same probability. A generalised extreme-value law fitted to the same draws [28] measures the tail's shape ξ: zero for Gumbel's, negative for a lighter tail, for which p_{G} overstates the probability, and positive for a heavier one.

The 35 searched binaries are accounted for together. In each draw of a null, the highest of the 35 binaries' summits, each in units of its own nulls' scatter, gives one draw of the survey's highest null, and each binary's z is set against these draws with a Gumbel law fitted in the same way. Beside it, each binary's own p_{G} is compounded over the 35 [29], $1 - (1 - p_{\mathrm{G}})^{35}$. The larger of the two, under the null that gives the larger, is the false-alarm probability across the survey. A companion is taken as detected when its own p_{G} reaches 5σ (2.9×10^{-7}), and the probability across the survey is quoted beside it.

3.6 Uncertainty and masses

The statistical uncertainty of K_2 is half the 16th to 84th percentile range of 300 draws of whole nights with replacement, each cleaned again and its peak found on grids of 0.25 and then $0.05\,\mathrm{km\,s^{-1}}$. Noise that persists over a season is not in it. As in the published analyses [13, 17], each series is divided by date into four parts of about equal numbers of spectra (two parts below 60 spectra), whole nights kept together, each part cleaned and measured alone, and the parts' standard error with Bessel's correction is a systematic uncertainty. The series is also resampled by observing season, runs of nights without a gap longer than 90 d, in 300 draws of whole seasons with replacement when it spans four seasons or more. The larger of the two is the systematic uncertainty, added in quadrature to the statistical one for the masses, which follow from

$$M_{1,2} \sin^3 i = 1.0361 \times 10^{-7}\,(1 - e^2)^{3/2}\,(K_1 + K_2)^2\,K_{2,1}\,P, \tag{5}$$

with the velocities in $\mathrm{km\,s^{-1}}$, the period in days and the masses in $\mathrm{M_\odot}$, divided by $\sin^3 i$ at the published inclination from the eclipses.

3.7 Robustness tests

Each companion is measured again under five changes of the analysis: three and eight principal components in place of five; the M-dwarf template kept only blueward or only redward of 5500 Å;

Table 1: The primary's orbits fitted to the velocities measured here, against the published orbits. ω is the primary's argument of periastron; rms is the scatter of the velocities about the fit.

System	Source	P (d)	K_1 ($\mathrm{km\,s^{-1}}$)	e	ω (°)	rms ($\mathrm{m\,s^{-1}}$)
TOI-1338	this work	14.608560	21.616	0.1556	117.86	14
	ref. [13]	14.6085579	21.604	0.1555	117.77	
J0310−31	this work	12.642791	27.868	0.3089	185.83	15
	ref. [17]	12.6427926	27.875	0.3087	185.85	
HD 4875	this work	13.635615	29.736	0.1814	251.71	29
	ref. [21]	13.6355611	29.739	0.1811	252.03	

and the template made instead from the 110 archived HARPS spectra of L 98-59, an M3 dwarf with the same instrument [26], in the same way as GJ 367's. The band and template changes rebuild the rows from the spectra. A synthetic companion is also laid into the actual spectra before any processing: the template's lines at a V-band flux ratio of 10^{-3}, and for the four companions of Section 5 also 3×10^{-4}, scaled in wavelength as the ratio of 3400 and 6000 K black bodies, moving at the measured K_2 and an offset of $+1.0\,\mathrm{km\,s^{-1}}$ on the primary's orbit but a quarter of a period later than the real companion, so that the two tracks cross only twice an orbit. The injected spectra go through the whole analysis and are read along the moved track, the null with the phase moved kept a tenth of a period from the real companion's track. The ratio of the real companion's peak to the injected one's at 10^{-3} gives the flux ratio at which the real companion's lines match the template's.

4 Three systems with published dynamical masses

Three of the binaries have published measurements of the secondary's semi-amplitude from other data or other methods: TOI-1338 (EBLM J0608−59), measured from ESPRESSO spectra with Saltire [13]; EBLM J0310−31, measured from HARPS spectra with Saltire [17]; and HD 4875 (EBLM J0050−18), measured from NIRPS spectra by cross-correlation with a synthetic M-dwarf spectrum [21]. The archive holds 264 ESPRESSO spectra of TOI-1338 above the signal-to-noise cut, of which 260 load, 114 HARPS spectra of J0310−31 and 35 of HD 4875. They test the method's accuracy, its uncertainties and the calibration of its false-alarm probabilities on companions whose semi-amplitudes are known.

4.1 Orbits

The orbits fitted to the measured velocities (Table 1) leave a scatter of 14, 15 and $29\,\mathrm{m\,s^{-1}}$, which is the precision of a cross-correlation velocity against a template built from one star's spectra, and they agree with the published orbits to within $0.012\,\mathrm{km\,s^{-1}}$ in K_1 and 0.0003 in eccentricity. The systemic velocities differ from the published ones by 0.01 to $0.14\,\mathrm{km\,s^{-1}}$, the offset between the template's zero point and those of the published analyses, which enters the companion's track only through Δv_{sys}.

4.2 The companions

The companion stands out in each system's map (Fig. 1) at the published K_2 and just redward of the primary's systemic velocity, and the trail over K_2 at that offset rises alone above the nulls

Table 2: The secondary's semi-amplitude, with its statistical and systematic uncertainties (Section 3.6), the companion's systemic offset with the 16th to 84th percentiles of 300 draws of whole nights, the summit at an offset of $+1.0\,\mathrm{km\,s^{-1}}$ in units of its nulls' scatter with its false-alarm probability (Section 3.5), and the masses at the published inclinations, 89.639°, 86.531° [17] and 89.057° [21], against the published values.

	TOI-1338	J0310−31	HD 4875
Spectra (nights), instrument	260 (255), ESPRESSO	114 (114), HARPS	35 (33), HARPS
Span (d)	2231	3031	1376
K_2 ($\mathrm{km\,s^{-1}}$)	$77.27 \pm 0.10 \pm 0.15$	$88.12 \pm 0.18 \pm 0.34$	$75.29 \pm 0.30 \pm 0.87$
Δv_{sys} ($\mathrm{km\,s^{-1}}$)	+1.05 (+1.00 to +1.15)	+1.10 (+1.00 to +1.25)	+1.25 (+0.75 to +1.25)
Summit in nulls' scatter, z	19.3	19.4	13.8
False-alarm probability	10^{-11} (6.7σ)	8.3×10^{-12} (6.7σ)	1.1×10^{-8} (5.6σ)
Across the 35 binaries	–	2.9×10^{-10} (6.2σ)	3.8×10^{-7} (4.9σ)
M_1 ($\mathrm{M_\odot}$)	1.102 ± 0.007	1.343 ± 0.015	1.116 ± 0.033
M_2 ($\mathrm{M_\odot}$)	0.308 ± 0.001	0.425 ± 0.003	0.441 ± 0.008
Published K_2 ($\mathrm{km\,s^{-1}}$)	77.16 ± 0.48	88.27 ± 0.46	75.13 ± 0.16
Published M_1 ($\mathrm{M_\odot}$)	1.098 ± 0.017	1.350 ± 0.017	1.1150 ± 0.0066
Published M_2 ($\mathrm{M_\odot}$)	0.307 ± 0.003	0.426 ± 0.003	0.4405 ± 0.0015
Spectra used	103, ESPRESSO	71, HARPS	15 nights, NIRPS
Reference	[13]	[17]	[21]

(Fig. 2). Table 2 sets the results against the published ones. The semi-amplitudes differ from the published values by +0.11, −0.15 and $+0.16\,\mathrm{km\,s^{-1}}$, which is 0.2, 0.3 and 0.2 times the two uncertainties combined, and at the published inclinations the masses agree within 0.6 per cent. HD 4875's published value comes from near-infrared spectra alone.

The summits stand 19.3, 19.4 and 13.8 times the nulls' scatter above their mean, where the highest of the adopted null's 10^4 draws reaches 3.1, 5.2 and 3.0 times. No draw comes within reach of the observed summits, so the empirical probabilities are at their floor of 10^{-4}, and the Gumbel law gives 10^{-11}, 8×10^{-12} and 1×10^{-8}, or 6.7, 6.7 and 5.6σ (Table 4). With the offset free within $5\,\mathrm{km\,s^{-1}}$ of zero they are 2×10^{-12}, 3×10^{-10} and 2×10^{-7}. Across the survey's 35 binaries, J0310−31 stays at 3×10^{-10} (6.2σ) and HD 4875 at 4×10^{-7} (4.9σ). The profile of the summed rows in the companion's frame (Fig. 3) is a single line 5 to $6\,\mathrm{km\,s^{-1}}$ wide at half maximum, and nothing beyond $15\,\mathrm{km\,s^{-1}}$ of it rises above three times the profile's scatter.

4.3 The companion's systemic offset

The companion's lines sit +1.05, +1.10 and $+1.25\,\mathrm{km\,s^{-1}}$ from the primary's systemic velocity in the three systems. The published analyses, which leave the secondary's systemic velocity free and use the pipelines' M2 line mask, found the same sign and size: $+1.40 \pm 0.06\,\mathrm{km\,s^{-1}}$ in the ESPRESSO spectra of TOI-1338 [13] and $+1.20 \pm 0.10\,\mathrm{km\,s^{-1}}$ for J0310−31 [17]. The offset collects the difference between the two templates' zero points, the primary's G0 template on the scale of its Gaia orbit and GJ 367's spectrum placed at its pipeline velocities, and the two stars' differences in gravitational redshift and convective blueshift, which for J0310−31's radii [17] amount to about $0.1\,\mathrm{km\,s^{-1}}$ in gravitational redshift. Held at zero instead, the offset moves the summit to 77.62, 88.37 and $74.74\,\mathrm{km\,s^{-1}}$, by +0.35, +0.25 and $-0.55\,\mathrm{km\,s^{-1}}$, since a constant velocity error of the track projects onto K_2 through the phases the spectra sample. At this precision the offset is a parameter of the fit, and its uncertainty is part of the bootstrap's.

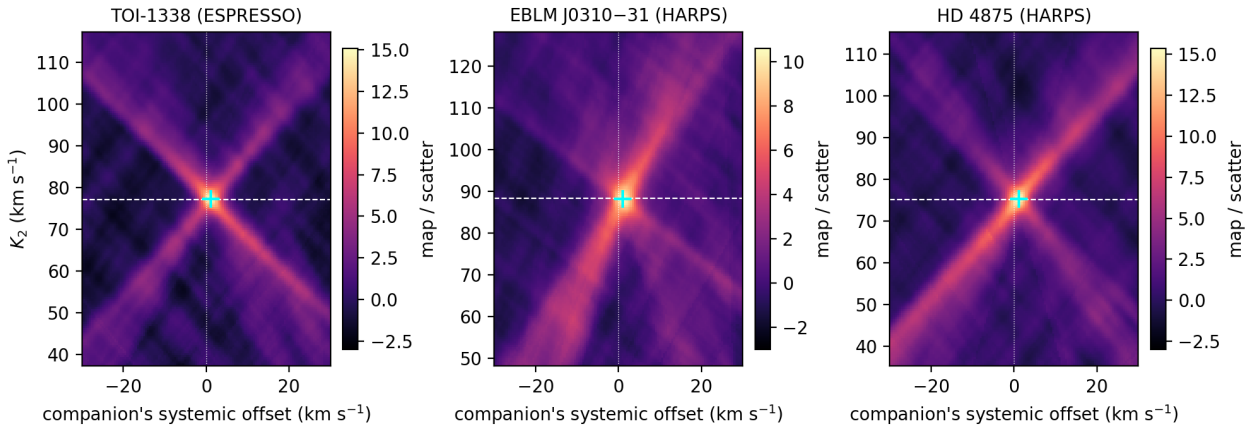


Figure 1: The mean of the cleaned rows along the companion's track over K_2 and the companion's systemic offset from the primary's, in units of the map's scatter away from the peak. The dashed line marks the published K_2, the dotted line zero offset and the cross the peak.

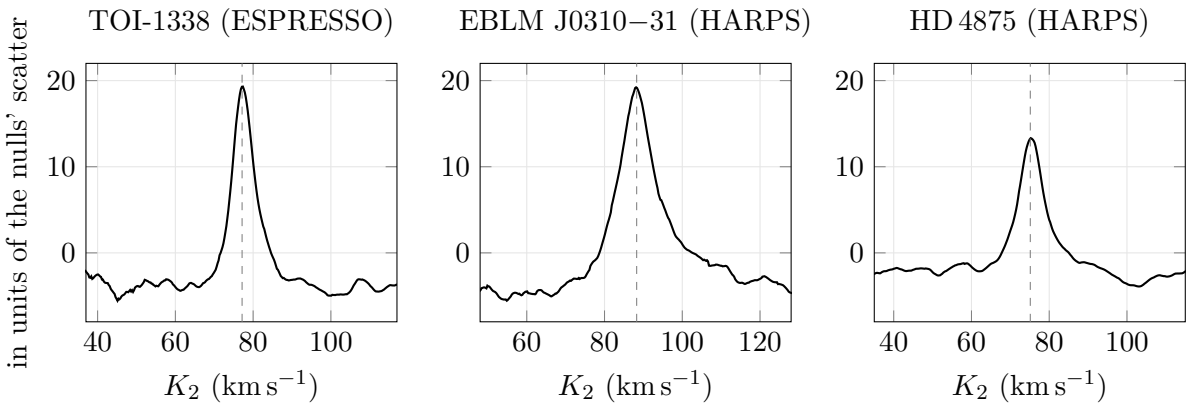


Figure 2: The trail over K_2 at the peak's systemic offset, less the mean of the stricter null's summits and in units of their scatter, from 300 draws of each null. The dashed line marks the published K_2.

4.4 Systematic uncertainty

Each series divided by date, and into alternate nights, gives the semi-amplitudes of Table 3. The four parts of TOI-1338 span 76.82 to 77.47 km s^{-1} and those of J0310−31 87.27 to 88.72 km s^{-1}, giving systematic uncertainties of 0.15 and 0.34 km s^{-1}, below the published uncertainties of 0.48 and 0.46 km s^{-1} that came from splits by observing season [13, 17]. HD 4875's 35 spectra divide into two halves, at 74.24 and 75.99 km s^{-1}, whose standard error of 0.87 km s^{-1} is three times the night bootstrap's 0.30 km s^{-1} and makes its total uncertainty 0.93 km s^{-1}. Resampling whole seasons gives smaller ranges for all three. The systematic term therefore dominates the uncertainty of every semi-amplitude measured here, as it did in the published analyses.

4.5 Robustness and injections

The five changes of the analysis (Section 3.7) leave the semi-amplitudes within 0.05 km s^{-1} of the measured value for TOI-1338, within 0.25 km s^{-1} for J0310−31 and within 0.40 km s^{-1} for HD 4875 (Table 7), each within its total uncertainty. Each half of the band finds every companion on its

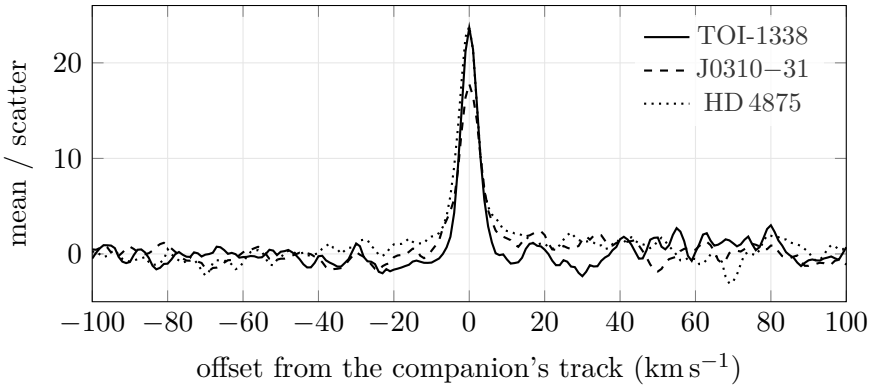


Figure 3: The cleaned rows summed in the companion's frame at the peak's K_2 and systemic offset, in units of the profile's scatter beyond 30 km s^{-1}.

Table 3: The semi-amplitude, with the summit's z in brackets, from each part of a series by date (two parts below 60 spectra) and from alternate nights, each cleaned and measured alone, the range from resampling whole observing seasons, and the systematic uncertainty, the larger of the parts' standard error and the seasons' half-range.

Subset	TOI-1338	J0310−31	HD 4875
Part 1 by date	76.82 (11.5)	87.52 (7.1)	74.24 (5.2)
Part 2 by date	77.47 (12.2)	87.27 (8.9)	75.99 (10.2)
Part 3 by date	77.37 (8.9)	88.32 (8.1)	–
Part 4 by date	77.42 (13.2)	88.72 (3.2)	–
Alternate nights, one set	77.02 (15.6)	87.97 (12.0)	75.34 (9.3)
Alternate nights, other set	77.17 (17.3)	87.97 (13.4)	75.19 (11.6)
Seasons resampled, 16th–84th	77.17–77.37	88.02–88.37	74.94–75.44
Systematic uncertainty (km s^{-1})	0.15	0.34	0.87

own, at 12 to 20σ in the units of the nulls' scatter, and the L 98-59 template finds it at least as strongly as GJ 367's. Companions injected at a flux ratio of 10^{-3} a quarter of a period from the real ones are recovered at -0.15, -0.10 and $+0.15$ km s^{-1} from the injected semi-amplitudes (Table 8), within the uncertainties and with no sign of a bias from the principal components the cleaning removes. Against the injections, the real companions' lines match the template's at flux ratios of 1.0, 0.9 and 2.9×10^{-3}.

4.6 Two HARPS series below the threshold

The 73 archived HARPS spectra of TOI-1338 do not show the companion on their own: the trail's summit over K_2 stands at $z = -0.1$, with a false-alarm probability of 0.49. Read at the published K_2 and the offset of $+1.0$ km s^{-1}, they give 2.8σ, with 21 of 2000 null draws as high, as the published analysis of HARPS spectra gave about 3σ [13]. For J1928−38, whose companion of 0.276 $M_\odot$ was measured at a signal-to-noise ratio of 3.65 in 73 HARPS spectra [17], the 74 archived HARPS spectra give no summit above the nulls and 0.3σ at the published K_2, with 718 of 2000 draws as high. At its brightness, $V = 11.2$, and its companion's mass, this method does not reach J1928−38's secondary in HARPS spectra.

Table 4: The false-alarm probabilities of the seven companions: the summit at an offset of $+1.0\,\mathrm{km\,s^{-1}}$ in units of its nulls' scatter (z), the highest of the adopted null's 10^4 draws in the same units, the Gumbel law's probability and its Gaussian significance, the same with the offset free within $5\,\mathrm{km\,s^{-1}}$ of zero, the probability across the survey's 35 binaries, and the fitted tail shape under the two nulls. No null draw reaches any of the seven summits, so their empirical probabilities are at the floor of 10^{-4}.

System	z	Highest null	p_G (σ)	Offset free	Across 35	ξ (phase, nights)
TOI-1338	19.3	3.1	10^{-11} (6.7)	1.6×10^{-12} (7.0)	–	0.02, 0.02
J0310−31	19.4	5.2	8.3×10^{-12} (6.7)	3.4×10^{-10} (6.2)	2.9×10^{-10} (6.2)	-0.10, -0.09
HD 4875	13.8	3.0	1.1×10^{-8} (5.6)	1.9×10^{-7} (5.1)	3.8×10^{-7} (4.9)	-0.01, -0.07
J2101−45	14.6	2.4	4.3×10^{-9} (5.8)	1.4×10^{-7} (5.1)	1.5×10^{-7} (5.1)	-0.34, -0.10
J0649−27	7.4	6.7	4.1×10^{-5} (3.9)	4.0×10^{-4} (3.4)	1.4×10^{-3} (3.0)	-0.22, -0.09
J0825−67	6.4	5.7	1.5×10^{-4} (3.6)	6.2×10^{-4} (3.2)	5.2×10^{-3} (2.6)	0.10, -0.02
J1141−37	5.4	4.9	5.7×10^{-4} (3.3)	3.1×10^{-3} (2.7)	0.02 (2.1)	-0.28, -0.09

5 The southern EBLM binaries

The same analysis was applied to the 37 EBLM binaries with archived HARPS series, with the companion's systemic offset held at $+1.0\,\mathrm{km\,s^{-1}}$, the value the three systems of Section 4 share, so that each search runs over K_2 alone. Two series cannot be searched: J2126−28 has 17 spectra above the signal-to-noise cut and J0543−57 has 21 spectra on six nights. Table 5 lists the other 35 with each summit's z and false-alarm probability.

5.1 Calibration across the survey

The nulls of the 35 binaries (Fig. 4) follow the Gumbel law to the depth 10^4 draws reach. The median of the 70 fitted shapes is $\xi = -0.02$, nine exceed $+0.1$ and the largest is $+0.81$, while the seven systems discussed here have ξ between -0.34 and $+0.10$ (Table 4). The survey's highest null, the largest of the 35 standardised summits of each draw, has a mean of 2.5 and 2.7 and a largest value in 10^4 draws of 5.1 and 8.5 under the two nulls. Four binaries reach 5σ on their own nulls: J0310−31 and HD 4875, J2101−45 at 4.3×10^{-9} (5.8σ), and TOI-1338's ESPRESSO series outside the survey. Across the survey J2101−45 keeps 1.5×10^{-7} (5.1σ), HD 4875 4×10^{-7} and J0310−31 3×10^{-10}. Three more stand between 3 and 5σ on their own nulls, J0649−27, J0825−67 and J1141−37 at 3.9, 3.6 and 3.3σ, but at 3.0, 2.6 and 2.1σ across the survey. The next highest, J0057−21, J0415−16 and J0621−50, have probabilities of 0.05, 0.08 and 0.37 across the survey, the last at the K_2 of $109.9\,\mathrm{km\,s^{-1}}$ that its model-based masses predict [4].

5.2 EBLM J2101−45

J2101−45's 75 spectra over 2298 d give $K_2 = 65.41 \pm 0.32 \pm 0.32\,\mathrm{km\,s^{-1}}$ at an offset of $+1.45\,\mathrm{km\,s^{-1}}$ (Table 6), the systematic term from its eight observing seasons. The companion holds under every change of the analysis (Table 7): the five variants give 65.26 to $65.91\,\mathrm{km\,s^{-1}}$, within 1.1 times the total uncertainty of the measured value, the red half of the band finds it at 21 and the blue half at 8 times the nulls' scatter, and the L 98-59 template at 19. Three of its four parts by date find it on their own, at 6.5 to 9.8 in the nulls' units, at 65.41 to $66.16\,\mathrm{km\,s^{-1}}$; the fourth, at 1.1, does not. A companion injected at a flux ratio of 10^{-3} is recovered at $65.61\,\mathrm{km\,s^{-1}}$, and against it the real companion's lines match the template's at 3.1×10^{-3}; one injected at 3×10^{-4} is not recovered. The masses are $m_1 \sin^3 i = 1.415 \pm 0.024$ and $m_2 \sin^3 i = 0.552 \pm 0.006\,\mathrm{M_\odot}$, and since

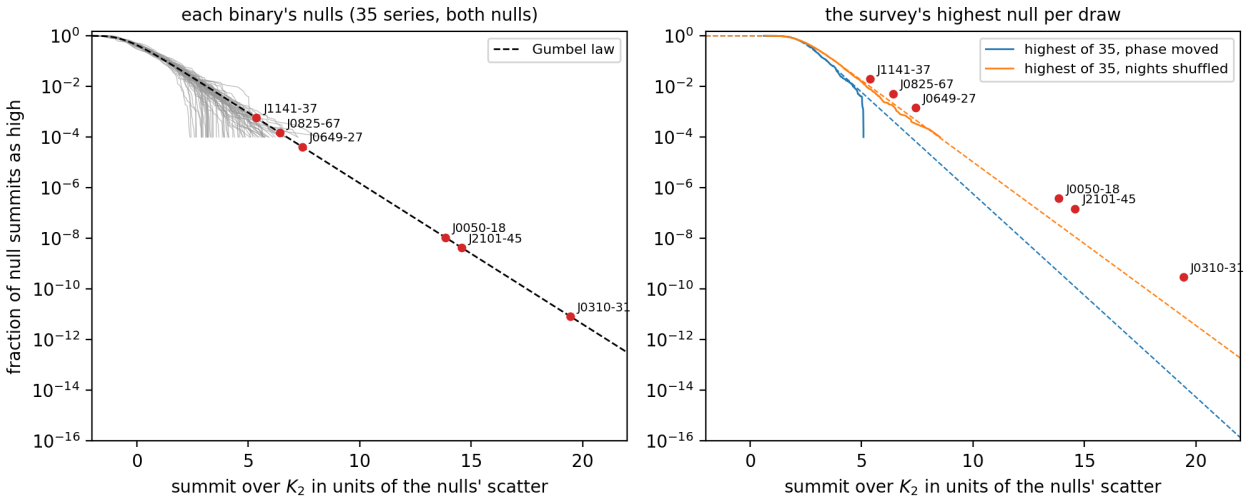


Figure 4: Left: the fraction of each binary's 2×10^4 null summits at or above a given height in units of their scatter, under both nulls, for the 35 binaries searched, against the Gumbel law of unit scatter (dashed), with the six companions of the survey at their observed z and p_{G}. Right: the same for the survey's highest null, the largest of the 35 standardised null summits in each draw, with the Gumbel laws fitted to it (dashed) and the six companions at their probabilities across the survey.

the binary eclipses, the primary's Gaia DR3 radius of $2.67\,\mathrm{R}_\odot$ requires $\sin^3 i > 0.993$, so the masses themselves are within 0.7 per cent of these. The primary, at $1.42\,\mathrm{M}_\odot$ and $2.67\,\mathrm{R}_\odot$, lies above the main sequence. The model-based masses of ref. [4], 1.29 ± 0.10 and $0.523 \pm 0.043\,\mathrm{M}_\odot$, differ from the dynamical ones by 1.2 and 0.7 times the combined uncertainties.

5.3 Three candidates

J0649−27, J0825−67 and J1141−37 do not reach 5σ on their own nulls and fall to 2.1 to 3.0σ across the survey, so their semi-amplitudes (Table 6, Fig. 5) are those their candidate companions would have. Each has a specific weakness. One of the 10^4 draws of J0649−27's adopted null reaches 6.7 times the nulls' scatter, against its summit's 7.4; neither half of its 43 spectra finds the companion on its own, at 1.0 and 1.7 in the nulls' units; and the red half of the band, where an M dwarf's lines carry most of the signal, peaks instead at $139.8\,\mathrm{km\,s^{-1}}$ and 2.8, while the blue half finds $120.35\,\mathrm{km\,s^{-1}}$ at 4.4. J0825−67 rests on 27 spectra, too few for its parts or alternate nights to be measured alone, and its uncertainty comes from its five seasons; companions injected along its moved track are not recovered at either flux ratio, the summit falling at $94\,\mathrm{km\,s^{-1}}$ and 0.7 to 0.9 in the nulls' units, so its spectra do not test the method's recovery. J1141−37's 30 spectra divide into halves at 1.1 and 3.5 in the nulls' units and alternate nights at 5.5 and 0.9, and its seasons give a systematic uncertainty of $2.6\,\mathrm{km\,s^{-1}}$. For J0649−27 and J1141−37 the injections at 10^{-3} are recovered within $0.15\,\mathrm{km\,s^{-1}}$, and against them the candidate companions' lines would match the template's at 0.7×10^{-3}, the order expected for an M3 dwarf beside an F star. The model-based masses of ref. [4] for J0649−27, 1.45 ± 0.13 and $0.371 \pm 0.061\,\mathrm{M}_\odot$, and for J1141−37, 1.22 ± 0.10 and $0.354 \pm 0.050\,\mathrm{M}_\odot$, are consistent with the minimum masses the candidates would give, 1.230 ± 0.071 and 0.335 ± 0.012, and 1.320 ± 0.090 and $0.371 \pm 0.015\,\mathrm{M}_\odot$, which the eclipses bound by $\sin^3 i > 0.96$. More spectra of these three would decide them.

Table 5: The 35 searchable EBLM binaries with archived HARPS series: the spectra (nights), the span, the primary's orbit fitted to the measured velocities, and the summit of the trail over K_2 at a systemic offset of $+1.0\,\mathrm{km\,s^{-1}}$ with its z and false-alarm probability. Companions at 5σ or more are in bold and candidates between 3 and 5σ in italics.

EBLM	Spectra	Span (d)	P (d)	K_1 ($\mathrm{km\,s^{-1}}$)	e	K_2 ($\mathrm{km\,s^{-1}}$)	z	p_G
J0035−69	27 (27)	949	8.4147	17.33	0.244	98.33	0.1	0.38
J0050−18	35 (33)	1376	13.6356	29.74	0.181	**75.24**	**13.8**	1.1×10^{-8}
J0055−00	83 (80)	2771	11.3918	21.16	0.058	118.91	-0.9	0.85
J0057−21	35 (35)	2288	32.3014	17.87	0.220	63.37	4.6	1.5×10^{-3}
J0104−38	35 (35)	1910	8.2561	20.65	0.000	100.40	2.1	0.04
J0110−40	43 (43)	1740	28.7816	10.23	0.092	26.98	-1.0	0.87
J0310−31	114 (114)	3031	12.6428	27.87	0.309	**88.37**	**19.4**	8.3×10^{-12}
J0353−16	120 (100)	3960	11.7612	16.64	0.006	83.14	1.2	0.11
J0415−16	48 (47)	1390	51.7026	15.21	0.239	50.46	4.3	2.4×10^{-3}
J0418−53	91 (91)	2909	13.6993	10.36	0.000	109.11	-0.7	0.74
J0503−25	40 (39)	1968	15.2652	7.80	0.091	78.80	0.5	0.26
J0525−55	61 (61)	1885	4.8001	25.18	0.000	118.18	-0.3	0.57
J0526−34	85 (84)	2345	10.1909	23.60	0.126	94.60	2.5	0.02
J0608−59	73 (73)	1818	14.6085	21.62	0.156	77.62	-0.1	0.49
J0621−50	43 (43)	1778	4.9638	37.58	0.000	109.83	2.9	0.01
J0623−27	36 (36)	2493	5.7780	28.05	0.056	190.30	-0.4	0.60
J0627−67	41 (41)	1813	9.4689	17.27	0.159	42.27	-1.5	0.98
J0649−27	43 (42)	1797	4.3080	32.55	0.002	*119.55*	*7.4*	4.1×10^{-5}
J0759−69	81 (79)	1988	17.0043	16.26	0.413	23.51	0.2	0.34
J0825−67	27 (27)	1234	10.0669	29.43	0.052	*78.43*	*6.4*	1.5×10^{-4}
J0925−03	61 (61)	2168	14.7748	16.60	0.330	83.10	-0.2	0.54
J0946−29	39 (38)	1234	10.4511	20.82	0.161	56.82	-0.8	0.79
J0954−45	32 (32)	2262	8.0727	27.83	0.295	112.58	0.2	0.34
J1119−39	36 (36)	2257	4.2136	11.13	0.004	29.63	1.6	0.07
J1141−37	30 (30)	1718	5.1477	32.23	0.000	*116.23*	*5.4*	5.7×10^{-4}
J1201−36	65 (65)	2284	9.1131	8.74	0.153	28.99	-0.6	0.72
J1219−39	62 (62)	2175	6.7600	10.83	0.056	72.33	-1.3	0.95
J1305−31	41 (41)	2123	10.6191	22.40	0.038	82.65	0.0	0.43
J1613−76	39 (39)	1308	14.3087	22.90	0.068	82.40	0.4	0.29
J1630+10	37 (37)	1872	10.9638	19.44	0.183	84.94	-0.5	0.68
J1928−38	74 (73)	2271	23.3230	17.26	0.073	111.76	-1.4	0.96
J1959−56	40 (40)	1864	29.7492	17.71	0.365	100.71	-0.9	0.84
J2040−41	45 (44)	1901	14.4562	12.45	0.226	31.95	-0.9	0.82
J2101−45	75 (75)	2298	25.5768	25.51	0.091	**65.76**	**14.6**	4.3×10^{-9}
J2207−41	56 (55)	2308	14.7750	8.70	0.069	44.70	0.4	0.27

Table 6: J2101−45 and the three candidates, measured with the offset free: the semi-amplitude with its statistical and systematic uncertainties, the offset with the 16th to 84th percentiles of 300 draws of whole nights, the false-alarm probability with its Gaussian significance on the binary's own nulls and across the survey, and the minimum masses with the total uncertainty of K_2 carried through.

EBLM	K_2 (km s^{-1})	$\Delta v_{\rm sys}$ (km s^{-1})	$p_{\rm G}$ (σ)	Across 35 (σ)	$m_1 \sin^3 i$ (M$_\odot$)	$m_2 \sin^3 i$ (M$_\odot$)
J2101−45	65.41 ± 0.32 ± 0.32	+1.45 (+1.10, +1.65)	4.3×10^{-9} (5.8)	1.5×10^{-7} (5.1)	1.415 ± 0.024	0.552 ± 0.006
J0649−27	119.40 ± 2.02 ± 1.77	+1.35 (−0.67, +2.21)	4.1×10^{-5} (3.9)	1.4×10^{-3} (3.0)	1.230 ± 0.071	0.335 ± 0.012
J0825−67	78.18 ± 0.88 ± 0.75	+1.25 (+0.45, +2.05)	1.5×10^{-4} (3.6)	5.2×10^{-3} (2.6)	0.940 ± 0.034	0.354 ± 0.008
J1141−37	114.68 ± 1.65 ± 2.57	+0.15 (−0.40, +1.70)	5.7×10^{-4} (3.3)	0.02 (2.1)	1.320 ± 0.090	0.371 ± 0.015

Table 7: The semi-amplitude, with the summit's z in brackets, as measured and under the five changes of the analysis of Section 3.7: three and eight principal components, the template kept blueward or redward of 5500 Å, and the template from L 98-59's spectra.

System	As measured	3 components	8 components	3800–5500 Å	5500–7880 Å	L 98-59
TOI-1338	77.27 (19.3)	77.27 (21.1)	77.27 (19.0)	77.32 (12.8)	77.27 (19.8)	77.22 (21.7)
J0310−31	88.12 (19.2)	88.07 (19.7)	88.17 (19.8)	88.07 (13.5)	88.37 (12.1)	88.32 (20.8)
HD 4875	75.29 (13.3)	75.14 (11.1)	75.29 (14.7)	75.29 (15.6)	74.89 (15.5)	75.19 (14.8)
J2101−45	65.41 (15.6)	65.76 (12.2)	65.91 (15.8)	65.91 (8.4)	65.66 (21.2)	65.26 (19.0)
J0649−27	119.40 (6.8)	119.85 (6.9)	119.15 (5.5)	120.35 (4.4)	139.80 (2.8)	119.55 (6.8)
J0825−67	78.18 (6.6)	78.18 (6.1)	77.93 (7.4)	78.48 (3.0)	78.18 (8.2)	78.58 (7.0)
J1141−37	114.68 (5.5)	114.68 (6.0)	114.68 (5.4)	115.63 (4.2)	114.68 (5.3)	116.48 (5.1)

Table 8: Companions injected into the spectra a quarter of a period from the real ones: the flux ratio and K_2 injected, the semi-amplitude and offset recovered along the moved track with the summit's z, and, where the injection at 10^{-3} is recovered, the flux ratio at which the real companion's lines match the template's (10^{-3}).

System	Flux ratio	K_2 injected	K_2 recovered	$\Delta v_{\rm sys}$ (km s^{-1})	z	Real companion
TOI-1338	10^{-3}	77.27	77.12	+0.90	11.5	1.0
J0310−31	10^{-3}	88.12	88.02	+1.10	17.9	0.9
HD 4875	10^{-3}	75.29	75.44	+1.05	3.3	2.9
J2101−45	10^{-3}	65.41	65.61	+0.85	3.8	3.1
J2101−45	3×10^{-4}	65.41	26.76	−6.10	4.0	–
J0649−27	10^{-3}	119.40	119.45	+1.30	10.2	0.7
J0649−27	3×10^{-4}	119.40	38.70	−1.15	2.6	–
J0825−67	10^{-3}	78.18	94.13	+2.35	0.7	–
J0825−67	3×10^{-4}	78.18	94.23	+2.25	0.9	–
J1141−37	10^{-3}	114.68	114.83	+1.10	8.8	0.7
J1141−37	3×10^{-4}	114.68	115.03	+1.10	1.8	–

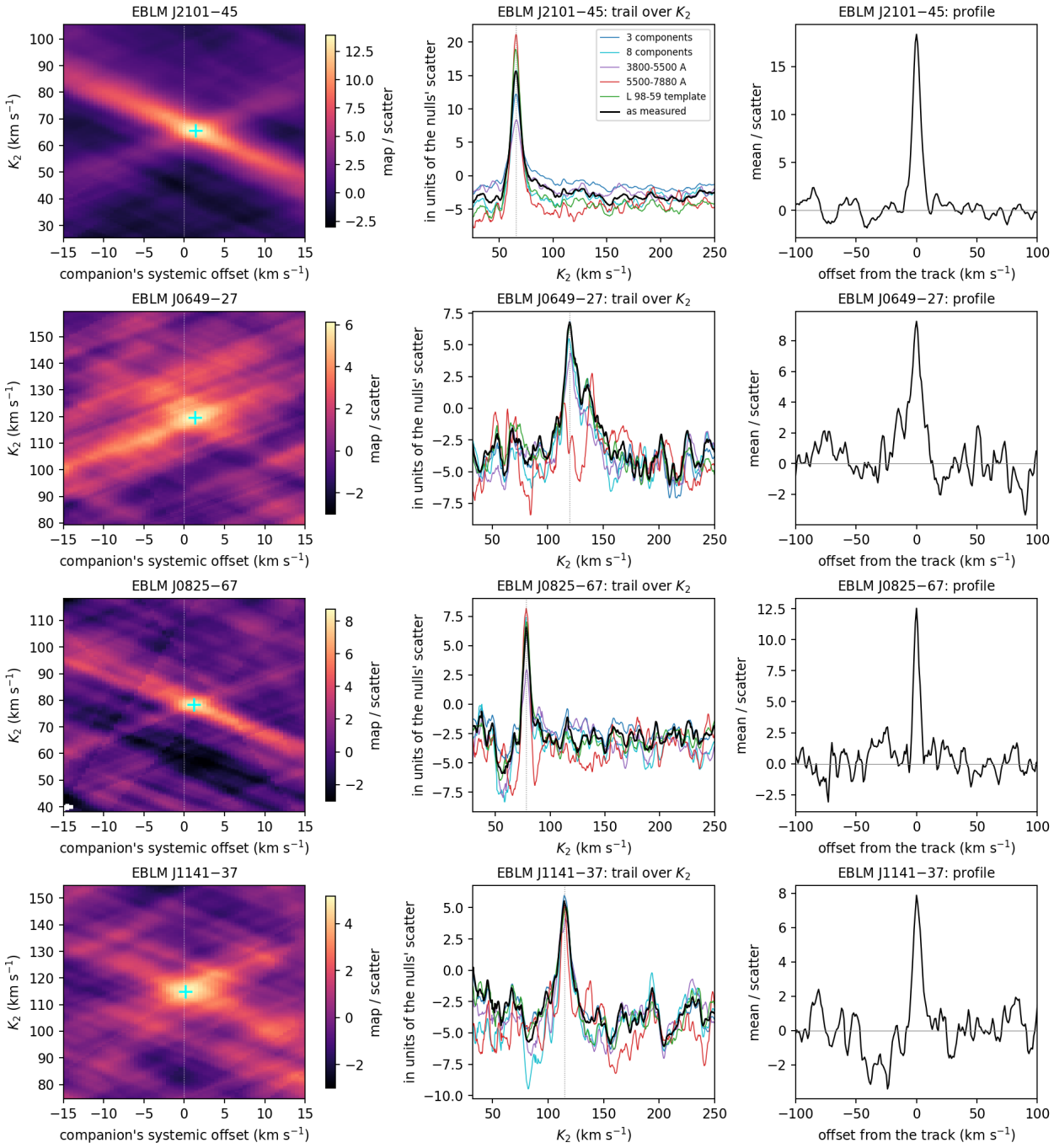


Figure 5: J2101−45 and the three candidates. Left: the map over K_2 and the companion's systemic offset in units of its scatter, the cross at the peak. Middle: the trail over K_2 at the peak's offset in units of the stricter null's scatter, as measured (black) and under the five changes of the analysis. Right: the cleaned rows summed in the companion's frame at the peak, in units of the profile's scatter beyond 30 km s^{-1}.

6 Discussion

The semi-amplitudes measured here differ from the published ones by -0.15 to $+0.16\,\mathrm{km\,s^{-1}}$, within the published uncertainties, although the analysis makes its own choice at each step. The template is an observed M1 V spectrum rather than a line mask or a synthetic spectrum; the primary's spectrum is removed by division by the series' median in the primary's frame and by principal components of the cross-correlation rows, rather than by a singular value decomposition of the spectra; the significance is measured against nulls built from the same rows rather than from a fit to the map; and the uncertainty comes from resampling whole nights and dividing the series by date. The agreement indicates that the published dynamical masses of TOI-1338 and EBLM J0310−31 do not depend on those analysis choices at the level of their quoted uncertainties, and that the near-infrared semi-amplitude of HD 4875's companion holds in the optical.

The calibration separates the height of a summit from its probability. In units of the nulls' scatter the four companions stand 13.8 to 19.4 high, but the nulls are drawn 10^4 times each, so the empirical probability stops at 10^{-4}, and beyond it the probability rests on the Gumbel law. That law fits the nulls to the depth they reach, and for the four companions the fitted tails are as light as Gumbel's or lighter, so their probabilities are not optimistic. Their summits also stand 11 to 16 times the nulls' scatter beyond the highest of the adopted null's 10^4 draws. Across the survey the probability of the weakest of the four, HD 4875, is 4×10^{-7}, and for J2101−45 1.5×10^{-7}. The three candidates show why the survey-wide accounting matters: at 3.3 to 3.9σ alone they would each look like detections, and 35 binaries searched over K_2 make summits of that height expected.

The archive's own completeness matters as much as the method. It holds 114 HARPS spectra of J0310−31 above a signal-to-noise ratio of 10, against the 71 of the published analysis, and 260 usable ESPRESSO spectra of TOI-1338, two and a half times the published set. A detection's significance grows with the square root of the number of spectra at fixed quality, and the candidates, with 27 to 43 spectra, are where more spectra would count.

The method has the limits of any optical analysis of these binaries. Toward $0.1\,\mathrm{M_\odot}$ the companion's optical flux falls to 10^{-4} of the primary's and below, and the HARPS series of J1928−38 and of the lowest-mass companions do not reach them; no injection at 3×10^{-4} is detected, and only J1141−37's returns the injected K_2, at 1.8 in the nulls' units. The near-infrared carries a larger share of the companion's light, and the archive holds NIRPS series of nine of the EBLM binaries considered here, taken alongside their HARPS spectra.

7 Conclusion

An analysis of public ESO archive spectra, independent of the published pipelines, recovers the M-dwarf companions of TOI-1338, EBLM J0310−31 and HD 4875 with false-alarm probabilities of 10^{-11}, 8×10^{-12} and 1×10^{-8} and measures their semi-amplitudes as 77.27 ± 0.18, 88.12 ± 0.38 and $75.29 \pm 0.93\,\mathrm{km\,s^{-1}}$, within $0.16\,\mathrm{km\,s^{-1}}$ of the published values, with masses within 0.6 per cent of the published dynamical masses. HD 4875's companion, measured before in the near-infrared alone, is detected here in the optical. The companion's lines sit $+1.05$ to $+1.25\,\mathrm{km\,s^{-1}}$ from the primary's systemic velocity in all three, and leaving that offset free is needed for accuracy at the tenth of a kilometre per second. Across the 35 searchable southern EBLM binaries with archived HARPS series, the companion of EBLM J2101−45 is measured for the first time, at a false-alarm probability of 1.5×10^{-7} across the survey, with $K_2 = 65.41 \pm 0.46\,\mathrm{km\,s^{-1}}$ and minimum masses of 1.415 and $0.552\,\mathrm{M_\odot}$; the companions of J0649−27, J0825−67 and J1141−37 remain candidates.

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Data availability

The spectra are public in the ESO Science Archive. The analysis is the binaries approach of the seti-tools package that holds this paper (`seti.py composition binaries`, whose modes build the rows, search, calibrate and measure), and its coverage store records the run as `binaries_eblm_known_orbits`. The paper's data folder holds the measurements and their maps, curves and profiles, the robustness and injection measurements, the calibrations, the survey's search results, and the tables written from them.

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